Statement of Purpose – Hafiz Adjei

Growing up in Ghana, I observed how paramount agriculture is to daily survival and economic stability. I also saw how fragile it can be. Farmers in my community located in the Northern part of Ghana relied heavily on instinct and historical weather patterns, leaving them vulnerable to unpredictable climate shifts and inadequate infrastructure. I realized early on that without localized, reliable data, sustainable farming in rural areas would always be a gamble. This realization drove my academic path in electrical engineering and ultimately shaped my focus on agricultural sensing. I am applying to the Precision Agriculture Lab at Southern Illinois University Carbondale because my background in building resilient, field-ready sensing systems uniquely positions me to help transition from laboratory-based predictive models into real-time, on-field applications.

My desire to bridge the gap between advanced hardware and rural farming culminated in my capstone project at Ashesi University. I designed and engineered a ground robot specifically for greenhouse monitoring. The objective was to collect spatial data on temperature, humidity, CO_2, and light intensity right at the crop level, without disrupting the farmers' daily routines. Building a UGV to navigate the tight, unpredictable rows of a greenhouse and maintaining reliable wireless transmission was an immense engineering challenge. I developed obstacle-avoidance algorithms, integrated the sensor payload, and built a lightweight remote dashboard. Watching the system successfully map environmental metrics in real time solidified my commitment to agricultural robotics and multi-sensor data collection.

However, collecting data in a controlled greenhouse is very different from deploying sensors in rural fields. Cloud-dependent IoT systems fail where internet access and power grids are non-existent. To solve this, I developed an environmental monitoring system utilizing a peer-to-peer LoRa network. By optimizing the hardware for extreme low-power consumption, I established a continuous, off-grid data pipeline that could monitor microclimates indefinitely. This work taught me the realities of field experimentation and the critical importance of hardware-software co-design in precision agriculture.

My recent experience as a Research Associate at CMU Africa has broadened my expertise in data integration and machine learning at the edge. I have spent last year reverse-engineering communication protocols and building full-stack monitoring systems using Python. Processing these complex data streams has prepared me for the large spectral and image datasets generated in your lab. I read your recent 2026 publication, *"AI-Augmented hyperspectral soil sensing: predictive modeling of nitrogen and phosphorus using neural architecture search,"* with great interest. While your application of NAS to optimize CNN architectures proved highly effective for laboratory-based soil analysis, I was particularly drawn to your conclusion detailing the need to transition to on-field spectral data collection for precision decision-making.

This is exactly where I want to contribute. I want to explore how self-sustaining edge devices and UGV platforms can run lightweight, optimized machine learning models locally. By shifting the computational burden away from the cloud, we can perform real-time agronomic diagnostics directly in the field. I bring a proven ability to conduct independent hardware research and a profound personal commitment to the future of farming. I look forward to the opportunity to bring my field robotics expertise to the Precision Agriculture Lab.

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